

FlashForge Dreamer OrcaSlicer profile guide

Original firmware | OrcaSlicer 2.4.2 | Validated T0/PLA and T1/PETG

Source profile

These bundles were translated from:

- File: `D:\3dPrintModels\FlashForge Dreamer.fff`
- Simplify3D profile: **FlashForge Dreamer**
- Simplify3D version marker: **S3D-Software 3.0.0**
- File modified: **9 May 2020**
- SHA-256: `8715810A3D3770FADA422883D30EBC65AD1C691E9FFD8066E709F9BB5855E2CF`

The active process in the file is **ABS / Medium / Right Extruder Only**, but the FFF also contains complete **Left Extruder Only** and **Both Extruders** configurations plus PLA, ABS, PVA, Nylon, and PLA/PVA material presets.

Bundles

Bundle	Intended use	Status
<code>FlashForge_Dreamer_Right_T0_OrcaSlicer_2.4.orca_printer</code>	Normal printing and temperature towers with the right nozzle	Recommended
<code>FlashForge_Dreamer_Left_T1_OrcaSlicer_2.4.orca_printer</code>	Single-material printing with the left nozzle	Recommended; physically validated with PETG
<code>FlashForge_Dreamer_Dual_T0_T1_OrcaSlicer_2.4_EXPERIMENTAL.orca_printer</code>	Two-nozzle printing	Experimental; tool changes require hardware validation

Import with **File** → **Import** → **Preset Configs** or **Import Configs**. Orca documents `.orca_printer` as its complete printer bundle format: https://github.com/OrcaSlicer/OrcaSlicer/wiki/import_export

For the original temperature-tower task, import and select the **Right T0** bundle.

Machine mapping

Simplify3D FFF	OrcaSlicer	Translation
strokeXoverride=230	Printable X	230 mm
strokeYoverride=150	Printable Y	150 mm
strokeZoverride=140	Printable height	140 mm
Origin offsets 115,75,0	Printable polygon	X -115...115, Y -75...75
firmwareTypeOverride=FlashForge Dreamer Firmware	G-code flavour	Marlin (legacy) plus custom Dreamer G-code
Right tool 0	Right printer preset	M108 T0
Left tool 1	Left printer preset	M108 T1
toolheadOffsets=0,0 34,0	Firmware-owned nozzle offset	Orca offsets remain zero; M612 X34.5 Y0.0 is preserved
applyToolheadOffsets=0	Slicer must not apply offset	Prevents applying the 34.5 mm offset twice
Nozzle diameter 0.4	Nozzle diameter	0.4 mm
Filament diameter 1.75	Filament diameter	1.75 mm
relativeEdistances=0	Relative E disabled	Start G-code emits M82
independentExtruderAxes=1	Separate physical T0/T1 tools	Represented in the dual preset
Baud rate 115200	Not configured by Orca bundle	Relevant only to a separate serial uploader

FlashForge's own Dreamer Cura guide confirms the 230 × 150 × 140 mm volume and centre origin:

<https://en.fss.flashforge.com/10000/software/e329750d35ce3431fb8281e069ebec48.pdf>

Extrusion and retraction

FFF value	Orca value
Extrusion width	0.40 mm
First-layer width	125% = 0.50 mm
Sparse-infill width	200% = 0.80 mm
Retraction distance	1.0 mm
Retraction speed	1200 mm/min = 20 mm/s
Extra restart	0 mm
Z lift	0 mm
Wipe	Disabled
Coasting	Disabled; no direct Orca equivalent needed
Retract between layers	Enabled
Tool-change retraction, dual preset	12 mm at 600 mm/min = 10 mm/s
Tool-change restart adjustment	-0.5 mm

The FFF's "only retract when crossing outline" behaviour has no exact one-setting Orca equivalent. Retraction values are preserved, but Orca determines individual travel retractions using its own path planner.

Exact process translation

The FFF's original quality presets replace the provisional 0.16/0.28 mm presets:

Orca preset	Layer	First layer	Top/bottom	Infill	Support source value
0.10mm High	0.10 mm	0.09 mm	4/4	30%	40%
0.20mm Medium	0.20 mm	0.18 mm	3/3	20%	30%
0.30mm Fast	0.30 mm	0.27 mm	3/3	15%	25%

Speed conversion:

FFF setting	Calculation	Orca
Default speed	3600 mm/min	60 mm/s sparse infill
Outline underspeed	60×0.5	30 mm/s walls
Solid-infill underspeed	60×0.8	48 mm/s
Support underspeed	60×0.8	48 mm/s
Bridge speed multiplier	60×0.8	48 mm/s
First-layer underspeed	60×0.35	21 mm/s
Rapid XY	4200 mm/min	70 mm/s
Rapid Z	1200 mm/min	20 mm/s machine estimate

Other translated process settings:

- Two perimeters.
- Inside-to-outside wall ordering.
- Rectilinear sparse and solid infill.
- Infill angles 45°/–45°.
- 20% infill/wall overlap.
- Two skirt loops, 4 mm from the object.
- Support threshold 40°.
- Support XY separation 0.3 mm.
- Raft, prime pillar, and prime tower disabled.
- Arc fitting disabled because the original firmware profile did not request arcs.

Simplify3D's support percentage and grid-spacing model does not map one-to-one to Orca's support line-spacing model. The Orca presets use close approximations and should be treated as starting values when support is enabled.

Material translation

Material	Nozzle	Bed	Flow	Fan
PLA	205 °C first layer / 200 °C thereafter	60 °C	0.90	Off layer 1, 100% from layer 2
PETG	235 °C first layer / 230 °C thereafter	75 °C first layer / 70 °C thereafter	1.00	20–40%; off for first 3 layers
ABS	230 °C	105 °C	1.00	Off
PVA	195 °C	80 °C	1.00	Off layer 1, 100% from layer 2
Nylon/PA	220 °C	80 °C	1.00	Off
PLA right / PVA left	205/195 °C first layer; PLA 200 °C thereafter	70 °C	0.90/1.00	PLA updated from the validated tower; PVA remains source-derived

PETG is not present in the source FFF, but the added preset was physically calibrated on the left T1 nozzle using a 240–220 °C temperature tower. Visual inspection and an aggressive flex test selected 235 °C for the first layer and 230 °C thereafter, with a 75/70 °C bed. Use a very thin glue-stick film as a release barrier on the original blue build surface.

Start and end G-code

The source Dreamer commands are preserved:

```
GCODE
M146 r255 g0 b0
M907 X100 Y100 Z40 A100 B100
M132 X Y Z A B
M612 X34.5 Y0.0
M108 T0
M7
M6 T0
```

Orca temperature placeholders were added before the source sequence so the selected filament controls the actual targets. The source purge positions and feed rates are preserved.

The right-tool purge is at **Y=-74**; the left-tool purge is at **Y=-70**. Both run from approximately X -100 to 90 before wiping to X 110.

The ending sequence is now an exact translation of the FFF:

```
GCODE
M140 S0
M104 S0 T0
M104 S0 T1
G1 Z140 F3300
M107
G28 X0 Y0
M146 r0 g0 b0
M18
```

The Dreamer command reference describes **M108 T0/T1** as the required FlashForge tool selector; bare **T0/T1** should not be assumed to work: <https://github-wiki-see.page/m/Mrnt/OctoPrint-FlashForge/wiki/G-Code-Reference>

Dual-extruder caveat

The FFF supplies the dual start sequence and tool-change retraction values, but its **toolChangeGcode** field is empty because Simplify3D's Dreamer firmware driver handled tool selection internally.

For Orca, the dual bundle explicitly uses:

```
GCODE
M108 T[next_extruder]
G92 E0
```

That is the correct translation in principle, but it must be tested physically because Orca and Simplify3D do not schedule dual-tool paths identically. The dual preset deliberately leaves the firmware responsible for the 34.5 mm offset to avoid applying it twice.

Before a dual material print:

1. Perform a no-model or raised-Z air-print.
2. Confirm **M108 T0** drives the right extruder and **M108 T1** drives the left.
3. Confirm each purge line uses the intended nozzle.
4. Print a small two-colour alignment square.
5. Check the actual X/Y alignment before attempting a large model.

Temperature tower

1. Select **FlashForge Dreamer - Right T0 0.4 nozzle**.
2. Select **Dreamer Generic PLA - Right T0**.
3. Select **0.20mm Medium @Dreamer Right T0**.
4. Choose **Calibration** → **Temperature**.
5. A reasonable first PLA range is 220 °C down to 185 °C in 5 °C steps, provided it agrees with the spool's stated range.
6. Slice and colour the Preview by temperature.
7. Before printing, confirm the exported G-code contains:
 - `M108 T0`
 - `M82`
 - `M612 X34.5 Y0.0`
 - the expected `M104 S...` temperature steps
 - no `M108 T1` in a right-only tower

First-print checklist

- Level the bed using the Dreamer's normal procedure.
- Confirm Orca shows a centre-origin 230 × 150 mm bed.
- Confirm the loaded material and physical nozzle match the selected bundle.
- Ensure the front purge path is clear of glass clips.
- Start with a small air-print or calibration cube.
- Be ready to stop the printer during the first homing, purge, and Z-lowering sequence.
- Export G-code locally; the bundles do not configure Dreamer Wi-Fi or serial upload.
- If the printer does not list `.gcode`, rename a copy to the extension already used by the working Dreamer workflow, commonly `.g`.

Validation performed

Each `.orca_printer` archive was rebuilt against the JSON schema of the installed OrcaSlicer 2.4.2. Validation checks cover:

- Parseable JSON.
- Complete archive references.
- Exact printer compatibility strings for every filament and process preset.
- Expected one- or two-nozzle array counts.
- Centre-origin bed polygon.
- Absolute extrusion.
- Correct `M108` tool selection.
- `M612 X34.5 Y0.0`.
- Source-matched speed conversions.
- Source-matched shutdown sequence.

Physical print validation

The right T0 profile was then physically validated on the original-firmware Dreamer with a complete PLA temperature tower from 220 °C down to 185 °C. Heating, T0 selection, homing, purge, temperature changes, cooling, motion, and shutdown completed correctly. Visual comparison selected 200 °C as the best overall PLA temperature, with a 205 °C first layer and 60 °C bed.

The left T1 profile was physically validated with a PETG temperature tower from 240 °C down to 220 °C. T1 selection, inactive-T0 shutdown, heating, homing, purge, temperature changes, cooling, motion, and final shutdown completed correctly. The calibrated PETG preset uses 235 °C for the first layer, 230 °C thereafter, and a 75/70 °C bed. Dual-tool printing remains experimental and unvalidated.